

Introduction to Quadratic Functions

Objectives:

- Identify quadratic functions
- Determine how linear and quadratic functions are different.
- Sketch parabolas in standard form by using the vertex formula.

Common Core Standards:

F.IF.1 - Understand that a function from one set (called the domain) to another set (called the range) assigns to each element of the domain exactly one element of the range.

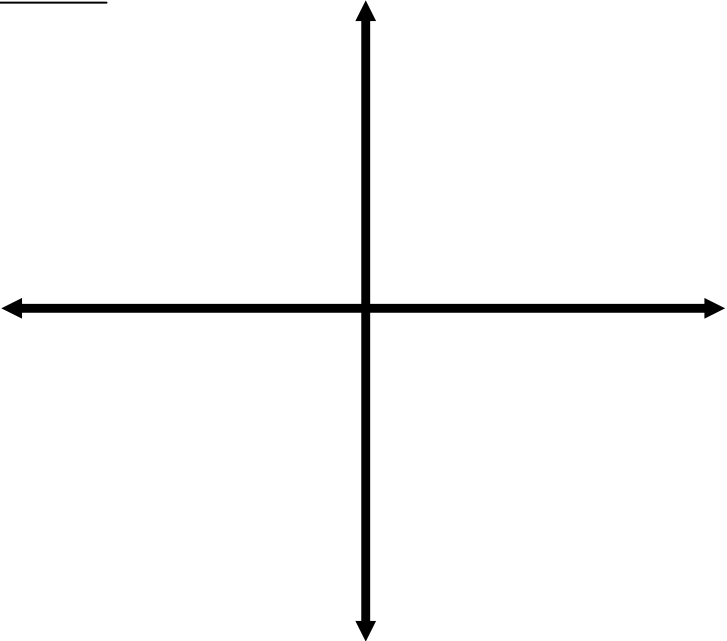
F.IF.4 - For a function that models a relationship between two quantities, interpret key features of graphs and tables in terms of the quantities, and sketch graphs showing key features given a verbal description of the relationship.

F.IF.7a - Graph functions linear and quadratic functions and show intercepts, maxima, and minima.

A.REI.10 - Understand that the graph of an equation in two variables is the set of all its solutions plotted in the coordinate plane. Often forming a curve (which could be a line).

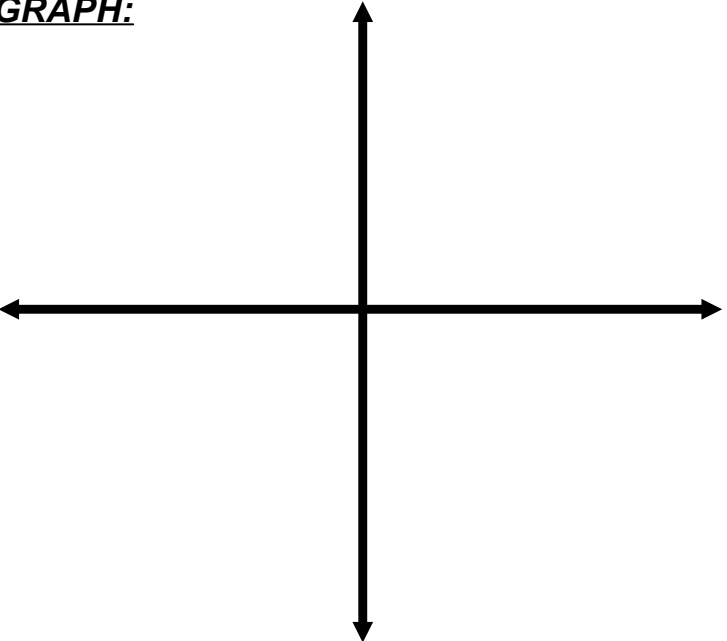
Linear Function

GRAPH:



Quadratic Function

GRAPH:



Standard Form of a Quadratic Function

Identify A, B, and C for the following quadratic functions.

$$y = 3x^2 + 5x + 2$$

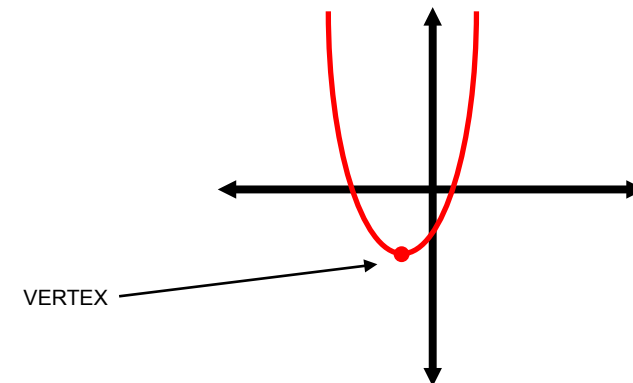
$$y = 6x - 7x^2$$

Vertex of a Parabola

The vertex will be the starting point when you sketch a parabola.

Vertex Formula:

Find the vertex of $y = 2x^2 + 8x + 4$.

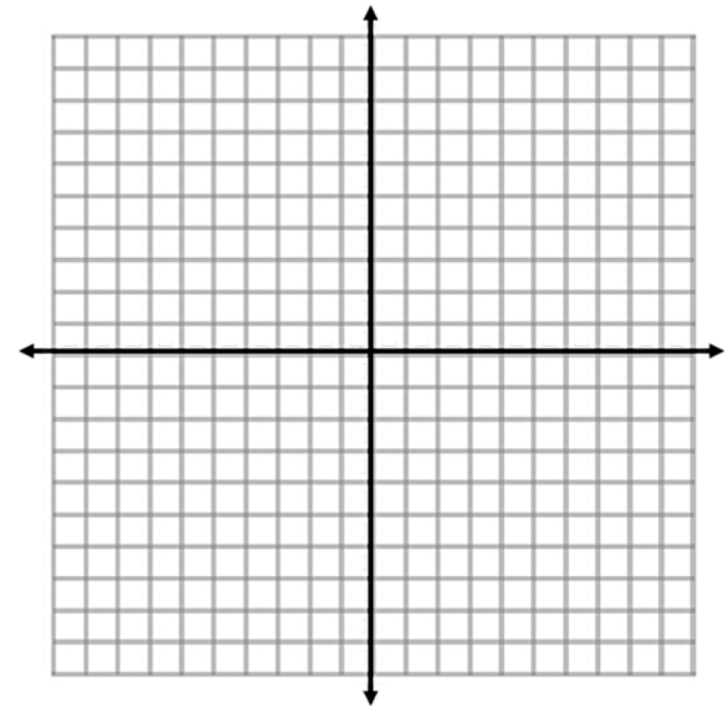


Graph $y = 2x^2 + 8x + 4$ on the coordinate plane.

- Find and graph the vertex.
- Use an x|y chart to find a point on each side of the vertex. (Pick an x-value on each side.)
- Use the points to sketch a parabola.

Vertex: $(-2, -4)$

x	$y = 2x^2 + 8x + 4$	y	(x, y)
-6	76	76	(-6, 76)
-5	49	49	(-5, 49)
-4	20	20	(-4, 20)
-3	4	4	(-3, 4)
-2	0	0	(-2, 0)
-1	2	2	(-1, 2)
0	4	4	(0, 4)
1	14	14	(1, 14)
2	28	28	(2, 28)
3	46	46	(3, 46)
4	68	68	(4, 68)
5	94	94	(5, 94)
6	124	124	(6, 124)

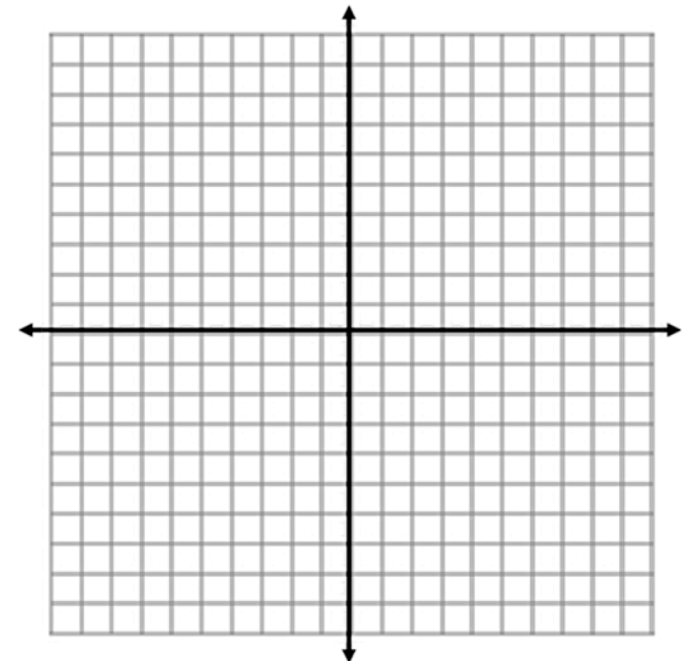


The vertex is the lowest point on the graph, so it is called the _____.

Graph $y = -3x^2$ on the coordinate plane.

- Find and graph the vertex.
- Use an x|y chart to find a point on each side of the vertex. (Pick an x-value on each side.)
- Use the points to sketch a parabola.

Vertex:

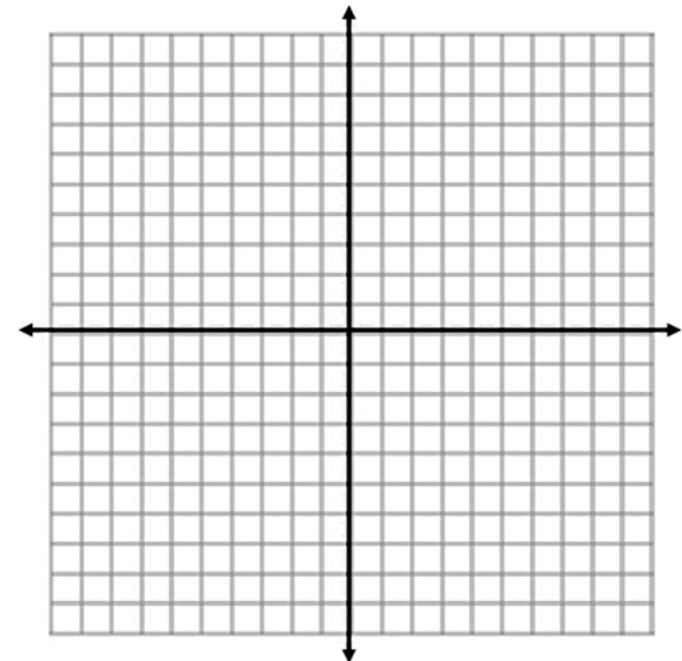
[illegible]

The vertex is the highest point on the graph, so it is called the _____.

Graph $y = x^2 + 2x$ on the coordinate plane.

- Find and graph the vertex.
- Use an x|y chart to find a point on each side of the vertex. (Pick an x-value on each side.)
- Use the points to sketch a parabola.

Vertex:

[illegible]

The vertex is the lowest point on the graph, so it is called the vertex.